

Customer Shopping Behaviour Analysis

1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900

- Columns: 18 - Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status) - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type) - Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.

```
[24]: import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')

[25]: df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	V
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	

- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```
[26]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   Customer ID           3900 non-null  int64  
 1   Age                   3900 non-null  int64  
 2   Gender                 3900 non-null  object  
 3   Item Purchased         3900 non-null  object  
 4   Category               3900 non-null  object  
 5   Purchase Amount (USD)  3900 non-null  int64  
 6   Location               3900 non-null  object  
 7   Size                   3900 non-null  object  
 8   Color                  3900 non-null  object  
 9   Season                 3900 non-null  object  
10  Review Rating          3863 non-null  float64 
11  Subscription Status     3900 non-null  object  
12  Shipping Type           3900 non-null  object  
13  Discount Applied        3900 non-null  object  
14  Promo Code Used         3900 non-null  object  
15  Previous Purchases      3900 non-null  int64  
16  Payment Method          3900 non-null  object  
17  Frequency of Purchases  3900 non-null  object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
[27]: df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

[28]:	df.isnull().sum()	[29]:	# Imputing missing values in Review Rating column with the median rating of the product category
[28]:	Customer ID 0		df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
	Age 0		
	Gender 0	[30]:	df.isnull().sum()
	Item Purchased 0	[30]:	Customer ID 0
	Category 0		Age 0
	Purchase Amount (USD) 0		Gender 0
	Location 0		Item Purchased 0
	Size 0		Category 0
	Color 0		Purchase Amount (USD) 0
	Season 0		Location 0
	Review Rating 37		Size 0
	Subscription Status 0		Color 0
	Shipping Type 0		Season 0
	Discount Applied 0		Review Rating 0
	Promo Code Used 0		Subscription Status 0
	Previous Purchases 0		Shipping Type 0
	Payment Method 0		Discount Applied 0
	Frequency of Purchases 0		Promo Code Used 0
	dtype: int64		Previous Purchases 0
			Payment Method 0
			Frequency of Purchases 0
			dtype: int64

- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created age_group column by binning customer ages.
 - Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender	revenue
►	Male	157890
	Female	75191

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	item_purchased	purchase_amount
►	2	Sweater	64
	3	Jeans	73
	4	Sandals	90
	7	Shirt	85
	9	Coat	97
	12	Shorts	68
	13	Coat	72
	16	Skirt	81
	20	Pants	90
	22	Pants	62
	24	Pants	88
	29	Handbag	94

3. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
►	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	round(avg(purchase_amount),2)
►	Express	60.48
	Standard	58.46

5. Subscribers vs Non-Subscribers – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
►	No	2847	59.87	170436
	Yes	1053	59.49	62645

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
►	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	number of customer
►	Loyal	3116
	Returning	701
	New	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
►	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

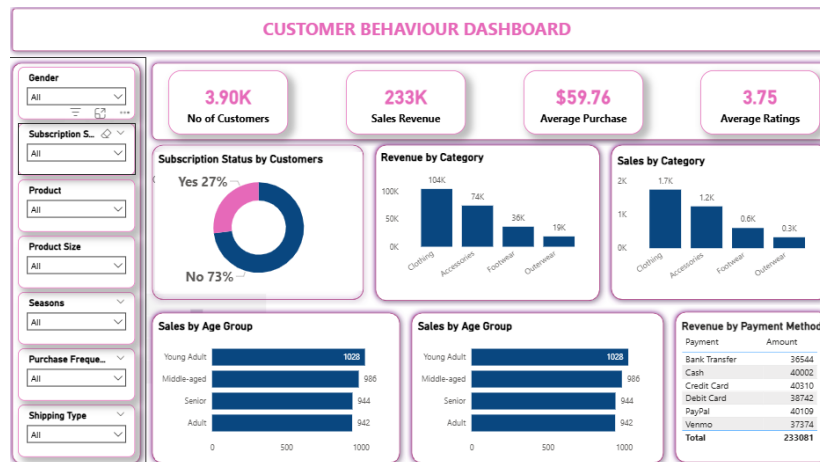
	subscription_status	repeat_buyers
►	Yes	958
	No	2518

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group	total_revenue
►	Young Adult	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.